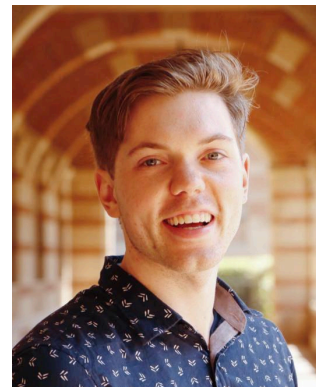


David Wyman Kanaar

10811 Ashton Ave, Apt 412 Los Angeles, CA 90024 tel:+1443 537 8830| dwkanaar@gmail.com



Research Summary

Postdoctoral researcher at UCLA specializing in the theory and simulation of semiconductor spin qubits, with expertise spanning many-body electronic structure (FCI), robust quantum control, and in-situ charge noise characterization. Seeking a research position in the Los Angeles area to continue advancing scalable quantum hardware.

Education

University of Maryland, Baltimore County (UMBC) | Baltimore, MD

- **Ph.D. in Physics** | *Graduated May 2024*
- Focus: Pulse shaping techniques for robust quantum gates.

Technische Universiteit Delft (TU Delft) | Delft, Netherlands

- **B.S. in Applied Physics** | *2015 – 2018*
- **Honours Track:** Additional research-focused curriculum.
- **Graduated Cum Laude**

University of Maryland, College Park (UMD) | College Park, MD

- **Study Abroad Program** | *2017*

Research Experience

UCLA – Gyure Group | *2024 – Present* **Postdoctoral Research Assistant** (Professors Mark Gyure & Chris Anderson)

- Utilizing **Full Configuration Interaction (FCI)** methods to investigate shell-filling and exchange properties in p-orbital (pO) qubits and EO qubits.
- Expended computational methods to calculate **valley splitting** in gate defined silicon spin devices.
- Developing advanced **quantum control** protocols and noise sensing techniques in solid-state systems.

UMBC – Kestner Group | *2020 – 2024* **Graduate Research Assistant** (Professor J.P. Kestner)

- Designed pulse-shaping **dynamical decoupling** techniques to create robust quantum gates and novel charge noise sensing methods.
- Utilized **neural networks** and Cartan decomposition for quantum control in silicon spin qubit arrays.

TU Delft – Steele Lab | *2018* **Research Assistant** Designed a 3D microwave cavity to measure Ferrimagnetic Resonance in Yttrium Iron Garnet (YIG).

TU Delft – Depken Lab | 2015 Research Intern Simulated virus DNA replication data to create and test predictive biological models.

Publications

- E. Eagan, **D. W. Kanaar**, J. Duan, J. M. Nichol, C. Anderson, and M. F. Gyure “Device simulation of p-orbital qubits on a realistic semiconductor architecture” (2026) in preparation
- **D. W. Kanaar**, J. Caporaletti, J. P. Kestner, C. Anderson, and M. F. Gyure “Validation of five electron *pO* qubit in silicon using FCI” in preparation (2026)
- R. Nodel, **D. W. Kanaar**, C. Nasseraddin, T. Wilson, H.W. Jiang, J. R. Petta, C. Anderson, and M. F. Gyure “Rapid Charge Stability Diagram Generation from Device-level Modeling of Semiconductor Quantum-dots” (2026) in preparation
- **D. W. Kanaar**, Y. Oda, M. F. Gyure, and J. P. Kestner, (2025), “Fast charge noise sensing using a spectator valley state in a singlet-triplet qubit,” *arXiv:2507.14108*.
- **D. W. Kanaar**, H. E. Ercan, M. F. Gyure, and J. P. Kestner, (2024), “Proposed real-time charge noise measurement via valley state reflectometry,” *arXiv:2402.14765*.
- **D. W. Kanaar** and J. P. Kestner, (2024), “Robust shaped pulses for arrays of superconducting or semiconductor spin qubits with fixed Ising coupling,” *Phys. Rev. A* 109, 032605.
- **D. W. Kanaar** and J. P. Kestner, (2024), “Neural-network-designed three-qubit gates robust against charge noise and crosstalk in silicon,” *Quantum Sci. Technol.* 9 035025.
- **D. W. Kanaar**, U. Güngördü, and J. P. Kestner, (2022), “Non-adiabatic quantum control of quantum dot arrays with fixed exchange using Cartan decomposition,” *Phil. Trans. R. Soc. A.* 380 20210275.
- **D. W. Kanaar**, U. Güngördü, and J. P. Kestner, (2022), “Two-qubit controlled-Z gates robust against charge noise in silicon while compensating for crosstalk using neural network,” *Phys. Rev. B* 105, 245308.
- **D. W. Kanaar**, S. Wolin, U. Güngördü, and J. P. Kestner, (2021), “Single-tone pulse sequences and robust two-tone shaped pulses for three silicon spin qubits with always-on exchange,” *Phys. Rev. B* 103, 235314.

Presentations

- **APS March Meeting 2026** | FCI shows *pO* qubits still work despite not following the shell-filling picture
- **Silicon Quantum Electronics Workshop (SiQEW) 2025** | Fast charge noise sensing in singlet-triplet qubits.
- **APS March Meeting 2025** | Spectator valley state sensing for singlet-triplet qubits.
- **IEEE Quantum Week 2024 (Montreal)** | *Invited Talk*: Designing silicon spin qubit gates robust to charge noise.
- **APS March Meeting 2024** | In-situ charge noise probe through the valley splitting.
- **APS March Meeting 2023** | Spin state transfer in fixed exchange quantum dot arrays using Cartan decomposition.
- **JHU APL Invited Talk (2022)** | Hamiltonian decomposition for robust quantum gates (Quiroz Group).
- **APS March Meeting 2021** | Pulse shaping for a robust CZ gate in silicon.

Professional Activities & Service

- **Conference session Chair:** APS March Meeting 2026 “Gate-Defined Spin Qubits: Control II”, APS March Meeting 2025 “Spin-Qubit Characterization and Modeling”
- **Peer Reviewer:** *Nature Communications Physics*, *Physical Review Applied and B*, *PRX Quantum*, *Scientific Reports*, *European Physics Letters*.
- **University Governance:** Graduate Member, UMBC Faculty Search Committees (2023, 2024).
- **Leadership:** Council Member, Physics Graduate Student Association (PGSA), UMBC.
- **Outreach:** Representative for UMBC at the Maryland UQA event (2023).
- **Membership:** American Physical Society (APS).

Teaching & Mentoring

- **Entangled Pairs Mentorship Program (UMBC):** Mentored undergraduate physics students (2023).
- **Teaching Assistant (UMBC):** PHYS 121 - Introductory Physics (2019 – 2020).
- **Aikido Instructor:** Taught lessons for children and adults (2012 – 2018).

Technical Competencies

- **Programming:** Python, Julia, MATLAB, C++.
- **Physics Software:** Mathematica, COMSOL, QuTiP.
- **Languages:** English (Fluent), Dutch (Fluent).
- **Specializations:** Device simulation, FCI calculations, Quantum control, Silicon spin qubits, Valley splitting physics, Charge noise mitigation